

Global Military Expenditure Analytics

Detailed Excel Data Cleaning, Data Modeling, Analysis & Dashboard Documentation

Project type: Data Analytics / Business Intelligence

Primary tool: Microsoft Excel — Power Query, Data Model, PivotTables, PivotCharts and Slicers

Deliverable: Interactive analytical dashboard with supporting data-model and analysis sheets

Project Purpose

This project transforms a raw global military expenditure dataset into a structured analytical workbook. The workflow covers data preparation, dimensional modeling, validation, exploratory analysis, KPI development, interactive visualization, and dashboard presentation. The final product answers practical questions about leading spenders, regional differences, historical trends and year-over-year changes.

1. Executive Summary

The source dataset contained approximately 198 rows and 79 columns and included real-world data quality issues such as nulls, empty cells, inconsistent country/region values, formatting differences and monetary values requiring normalization. Power Query was used to clean and standardize the source before it was loaded into Excel's analytical environment.

A simple dimensional model was created using country and year dimensions together with a military expenditure fact table. This supports reusable PivotTable analysis and consistent filtering. The final dashboard combines KPI cards, ranked country analysis, regional comparisons, historical trends, year-over-year growth and an interactive Year slicer.

2. Analytical Questions

- Which countries have the highest military expenditure?
- What is the total expenditure represented by the dataset?
- Which regions account for the largest expenditure?
- How has global military expenditure changed over time?
- Which years experienced the strongest increases or decreases?
- Can users interactively filter the analysis by year?

3. Source Data & Initial Assessment

The raw workbook contained 198 records and 79 columns. The dataset was inspected in Power Query before analysis. The initial assessment focused on nulls, blank values, inconsistent labels, unnecessary fields, data types, country/region mappings and monetary formatting.

4. Data Cleaning with Power Query

The cleaning workflow included importing the source, reviewing the 79 columns, identifying relevant analytical fields, checking null and blank values, standardizing country and region information, correcting the Africa-region inconsistency identified during validation, standardizing expenditure values, confirming numeric data types, and loading the cleaned data into Excel.

The guiding principle was to correct data-quality problems without changing the underlying meaning of the source records.

5. Data Model

The workbook follows a simple star-schema-style structure that separates descriptive attributes from measurable expenditure records.

Table	Role	Main fields / purpose
Dim_Country	Dimension	Country, CountryKey, Region
Dim_Year	Dimension	Year and year key/index
Fact_Military_Expenditure	Fact	Country, Year, Military Expenditure

CountryKey provides a stable country-level key and supports the relationship between the country dimension and fact table. Region is maintained in Dim_Country so regional reporting can be performed without duplicating regional attributes throughout the fact data.

6. Analysis Sheets

Top 10 Military Spenders

Ranks countries by total military expenditure and presents the highest-spending countries first. The chart is sorted in descending order so the largest spender appears at the top.

Regional Military Expenditure

Aggregates military expenditure by Region using the country dimension. The regional result is sorted from highest to lowest expenditure for direct comparison.

Military Expenditure Trend

Aggregates expenditure by year and visualizes the long-term global trend, highlighting sustained growth, declines and major changes.

Year-on-Year Expenditure Growth

Calculates percentage change from each year to the previous year. The first available year has no YoY value because there is no earlier comparison year.

7. Dashboard Design

The dashboard is the presentation layer of the project. It combines headline KPIs with four analytical visuals and an interactive Year slicer.

Component	Purpose
Total Military Expenditure	Headline expenditure measure
Years Covered	Shows analysis-period span
Top Spending Country	Leading country by expenditure
Highest Spending Region	Leading region
Top 10 Countries	Country ranking
Military Expenditure by Region	Regional comparison
Global Expenditure Trend	Historical time series
YoY Growth	Annual percentage change
Year Slicer	Interactive year filtering

The Year slicer was connected to the relevant PivotTables so users can change the selected year and update connected analysis views. Gridlines were removed, charts aligned, KPI presentation standardized and unnecessary legends removed.

8. Key Analytical Findings

- The United States is the highest-spending country in the dataset.
- North America is the highest-spending region in the completed regional aggregation.
- Global military expenditure shows a strong long-term upward pattern across the available historical period.
- YoY growth is volatile in some periods, with particularly large percentage changes during years with relatively low preceding-year expenditure.

These findings are specific to the supplied dataset and its defined period.

9. Data Validation & Quality Checks

- Country and region values were reviewed for consistency.
- The Africa region issue was corrected during transformation.
- Monetary fields were standardized to numeric values suitable for aggregation.
- Country keys were introduced for dimensional modeling.
- PivotTable totals and dashboard KPIs were checked against the underlying analysis.
- Charts were sorted and formatted to support correct visual ranking.
- The Year slicer was tested against connected analysis views.

10. Technical Skills Demonstrated

- Excel: PivotTables, PivotCharts, formulas, sorting, formatting and dashboard design
- Power Query: import, transformation, cleaning, standardization and data-type handling
- Data modeling: fact and dimension tables, keys and relationships
- Business intelligence: KPI design, ranking, trend analysis, regional analysis and interactive filtering
- Analytical thinking: converting a raw dataset into business-oriented questions and visual answers
- Data quality: identifying and resolving nulls, blanks, inconsistent classifications and formatting issues

11. Recommended Workbook Structure

1. Dashboard
2. Dim_Country
3. Dim_Year
4. Fact_Military_Expenditure
5. Top 10 Military Spenders
6. Regional Military Expenditure
7. Military Expenditure Trend
8. YoY Growth / YoY Calculation

12. GitHub Repository Recommendation

global-military-expenditure-analysis/

■■■ README.md

■■■ Global_Military_Expenditure_Analytics.xlsx

■■■ screenshots/

■ ■■■ dashboard.png

■■■ source/

■■■ Global_Military_Expenditure_Raw.xlsx

The README should explain the business questions, tools, data model, dashboard components and key insights. The raw dataset should be kept separate from the final analytical workbook.

13. Limitations & Future Improvements

Possible extensions include population-normalized spending, expenditure as a percentage of GDP, country-level trend drilldowns, regional share calculations, more advanced Power Pivot/DAX measures, and automated refresh from a stable data source. Conclusions depend on the quality, definitions, coverage and period of the source dataset.

14. Conclusion

This project demonstrates an end-to-end analytics workflow: starting with a messy multi-column dataset, cleaning and standardizing it, organizing the information into a dimensional model, building reusable analytical views, and communicating results through an interactive dashboard. The result is a portfolio-ready Excel project demonstrating both technical spreadsheet skills and structured analytical thinking.